

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	LSGI2BN01
Subject Title	Map Reading and Interpretation
Credit Value	3
Level	2
Exclusion	LSGI2B01 Map Reading and Interpretation
GUR Requirements Intended to Fulfill	Cluster Area Requirement (CAR) Cultures, Organisations, Societies and Globalisation “English Reading” (ER) designation - include a reading of an extensive text (100,000 words or 200 pages) “English Writing” (EW) designation - include an extensive piece of writing (1,500- 2,500 words)
Objectives	The powerful language of maps visually shows trends and patterns that are not apparent in other data presentations. Corporations, government, media, and researchers use maps and geographic information technology to understand and visualize data on, for example, natural resources, flows of trade, historical events, property management, and diseases. Students will explore what makes spatial information special, how and why maps are such a powerful tool to understand an increasingly complex world, and how modern technology is currently transforming the art and science of map-making. They will also learn how to be critical consumers of mapped information.
Intended Learning Outcomes	Upon completion of the subject, (a) Students will have read at least 200 pages of no less than 3 relevant reading sources. They will also be able to read, interpret, and understand different kinds of maps or spatially represented data. (b) Students will be able to critically assess how spatial data are represented through maps and recognize various issues. (c) Through active learning, students will acquire skills that enable them to understand the trends or facts represented in particular types of maps (e.g., transport, population, and environmental). (d) Produce a simple digital map for a particular theme of interest. Students are required to do an extensive reading from assigned textbooks or websites that will increase their literacy. Being able to read and interpret information from maps are essential and useful to many disciplines, everyday life, and life-long learning.

Subject Synopsis/ Indicative Syllabus	• Why is spatial special? Introduction to maps • Maps in Hong Kong • Map elements and types • Mapping through the ages • Mapping process – data collection and symbolization • Map evaluation • How to read and interpret maps? Maps of global and regional trends in transport, population, and environment.					
Teaching/Learning Methodology	<u>Lectures</u> - work with sample problems and discuss practical applications. These activities are meant to build a deeper understanding of the subject matter; active participation and preparation before classes are expected from students. <u>Tutorials</u> – students will get hands-on experience with the subject matter. Briefing and technical support for group projects.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Essay Writing*	40	✓			
	2. Assignment on map interpretation	10		✓	✓	
	3. Group project on thematic map	30		✓	✓	✓
	4. Test	20	✓	✓	✓	
	Total	100%				
*In order to pass the subject, students must pass the writing component assessment, i.e., attain a minimum grade D in the writing component based on at least 200 pages of no less than 3 relevant reading sources. Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The 40% essay writing is a requirement of an EWR subject in which students can have a thorough and in-depth understanding of the subject matter and be trained to express ideas critically. This is also reinforced by a test to check students’ understanding of the basic concepts in mapping. The group project enables students to work in a team and acquire the skills of producing a map digitally, from working out a meaningful theme and collecting relevant data to presenting data graphically and conveying the idea in a presentation.						
Student Study Effort Expected	Class contact:					
	▪ Lecture		26 Hrs.			
	▪ Tutorial		13 Hrs.			

	Other student study effort:	
	▪ ELC tutorial	2 Hrs.
	▪ Chapter (Book) Review	30 Hrs.
	▪ Group Projects	44 Hrs.
	▪ Self-study	10 Hrs.
	Total student study effort	125 Hrs.
Reading List and References	1. Crampton J. W. (2010) <i>Mapping A Critical Introduction to Cartography and GI</i>. Wiley-Blackwell. (hardcopy & electronic versions available in PolyU Library) 2. Kimerling, A.J., Muehrcke, J., Buckley, A. & Muehrcke, P. (2010) <i>Map Use: Reading and Analysis</i>, 6th Edition. 3. Krygier, J. & Wood, D. (2011) <i>Making Maps: A Visual Guide to Map Design for GIS</i> (2nd Edition). 4. Lemmens M. (2011) <i>Geo-information Technologies, Applications and the Environment</i>, Volume 5, Springer Dordrecht Heidelberg London New York. (electronic version) 5. Monmonier, M. (1996) <i>How to Lie With Maps</i>, 2nd Edition. 6. Robinson, A.H., Morrison, J.L., Muehrcke, P.C., Kimerling, A.J. & Guptill, S.C. (1995) <i>Elements of Cartography</i>, 6th Edition. 198 pages	